COGNITIVE DECISION INTELLIGENCE FRAMEWORK

For Dynamic Business Strategy Using Artificial Intelligence

Project Report — Editable Draft

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Abstract

This project proposes an AI-driven predictive analytics framework that brings together three business analytics use cases: Walmart weekly retail sales forecasting, Amazon customer-review sentiment classification, and M5 item-level demand forecasting. Dataset-specific preprocessing and machine-learning models are evaluated using task-appropriate metrics. A unified results layer is designed to feed an interactive Streamlit dashboard for model comparison and business decision support. This draft documents the intended workflow; measured results must be inserted after running the experiments on the authorized datasets.

1. Introduction

Organizations make decisions using information distributed across sales transactions, customer feedback, and demand histories. This project explores how machine-learning methods can support forecasting and customer understanding, while presenting model performance in a single decision-support interface. The framework is modular: each dataset has its own preparation and evaluation pipeline, and results are consolidated only after model execution.

2. Problem Statement

Business analysts often need to select models across different analytical tasks while communicating their performance to decision-makers. Separate notebooks and metrics can make this difficult. The project addresses the need for a reproducible workflow that prepares multiple datasets, trains suitable models, evaluates them with appropriate metrics, and presents the measured outputs in an accessible dashboard.

3. Objectives

- Prepare and explore Walmart, Amazon Reviews, and M5 data.
- Train and evaluate task-appropriate machine-learning models.
- Consolidate measured metrics into a consistent results schema.
- Build an interactive Streamlit interface for exploring results.
- Translate model findings into cautious, evidence-based business insights.

4. Dataset and Use Cases

Walmart: weekly store/department sales with store and calendar features; treated as a regression/forecasting task.
Amazon Reviews: review text and star ratings; a binary sentiment setup may map ratings 1–2 to negative and 4–5 to positive while excluding neutral 3-star reviews. Confirm the chosen policy against the source dataset.
M5: item/store daily sales with calendar and selling-price information; treated as a time-series demand forecasting task. Document any sampling used for compute limits.

5. Methodology

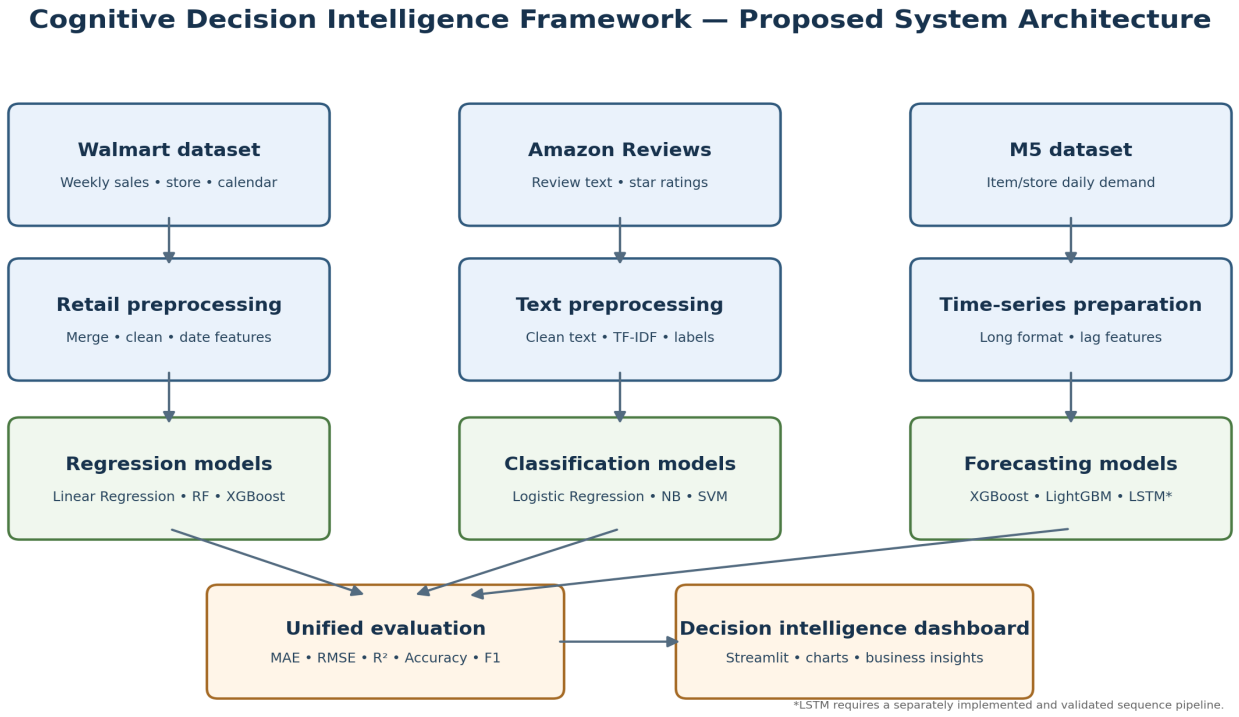
The workflow consists of data validation, dataset-specific preprocessing, feature engineering, chronological or stratified splitting as appropriate, model training, held-out evaluation, results consolidation, and dashboard visualization. Time-series experiments must preserve temporal order and ensure that lag and rolling features use past observations only. Text vectorization must be fitted within the training pipeline to avoid leakage.

6. Models and Evaluation

Walmart candidates: Linear Regression, Random Forest, and XGBoost. Metrics: MAE, RMSE, and R^2.
Amazon candidates: Logistic Regression, Multinomial Naive Bayes, and Linear SVM with TF-IDF features. Metrics: accuracy, precision, recall, F1-score, and confusion matrix.
M5 candidates: XGBoost, LightGBM, and an LSTM sequence model where implemented. Metrics: MAE and RMSE, evaluated on the same forecast horizon and validation dates.

7. System Architecture

The proposed architecture is shown below. The architecture is a design representation; update it if the final implementation differs.



8. Results and Discussion

Results are intentionally not populated in this draft. Insert measured metrics from results/unified_model_results.csv after executing the notebooks. For each dataset, identify the best model per metric, report the validation split and sample size, and discuss the practical significance. Do not compare unlike metrics across classification and forecasting tasks as if they were interchangeable.

9. Limitations

The framework's results depend on dataset version, sampling strategy, feature availability, hardware, and validation design. A small M5 sample is not equivalent to a full competition benchmark. A single holdout may not capture all seasonal regimes. Model explainability, robust backtesting, hyperparameter tuning, and deployment monitoring are potential extensions.

10. Future Work

Future work may include rolling-origin backtesting, hyperparameter optimization, SHAP-based explanation, a fully validated LSTM pipeline, experiment tracking, automated data refresh, model versioning, and deployment with monitoring and alerting.

11. Conclusion

This project establishes a modular approach for combining retail sales forecasting, review sentiment classification, and demand forecasting within a unified decision-support workflow. Its value should be demonstrated through reproducible experiments, transparent validation, measured model comparisons, and a working dashboard. Final conclusions must be updated using the actual experimental results.

References and Dataset Attribution

Add the exact dataset source citations, original dataset documentation, model/library references, and any relevant research papers used in the final implementation. Respect each dataset's license and redistribution terms.

Appendix: Finalization checklist

- Insert actual dataset versions and record counts.
- Add experimental environment and package versions.
- Add actual model scores and plots.
- Add genuine dashboard screenshots.
- Verify that all stated models were implemented and run.
- Add complete references and faculty/institution formatting as required.